BEN 10
CHARACTER UNIVERSE

Full Project Plan & Technical Blueprint

Public fan encyclopedia • AI-assisted research • Character and villain database

Included series
Ben 10 (2005)
Ben 10: Alien Force
Ben 10: Ultimate Alien

Unofficial fan project • Planning document

1. Executive Summary

Ben 10 Character Universe is a public, unofficial fan encyclopedia for exploring characters, aliens, transformations, heroes, villains, powers, weaknesses, appearances and galleries across three series. The site will combine a polished, responsive interface with a structured database and an AI-assisted research pipeline.

The AI agent will discover candidate characters, research facts, attach evidence, find candidate images and assess reuse terms. Content with uncertain facts or unclear image rights must remain unpublished until reviewed. The system should never invent missing canon details or treat an image being publicly accessible as permission to republish it.

Project goals

- Create a searchable character encyclopedia covering the three selected series.
- Give each character a dedicated profile with a target gallery of 3–4 suitable images when rights-cleared assets are available.
- Document powers, weaknesses, species, homeworld, relationships, forms and series appearances with sources.
- Automate repetitive research and data preparation while retaining editorial control over uncertain content.
- Build a maintainable architecture that can be expanded without rewriting the website.

2. Scope and Boundaries

Included series

Series	Coverage
Ben 10 (2005)	Original series characters, aliens, allies, villains and relevant forms.
Ben 10: Alien Force	Series-specific character versions, aliens, allies, villains and story-related details.
Ben 10: Ultimate Alien	Ultimate forms, new and returning characters, villains and series-specific abilities.

Explicitly excluded

- Ben 10: Omniverse — no dedicated series section or Omniverse-only content scope.
- Ben 10 (2016 reboot) — no reboot section or reboot-only content scope.
- Do not merge continuity-specific versions into a single undifferentiated record.
- Do not add unverified fan theories as canon facts.

Core content categories

- Aliens and species; human characters and allies; villains and antagonists.
- Transformations and Ultimate forms relevant to the included series.
- Character relationships, affiliations, homeworlds, appearances and documented abilities.
- Image galleries with source, creator/rights holder where known, license evidence, attribution and review status.

3. Target Users and Main User Journeys

User type	Main needs	Typical journey
Casual fan	Find a character quickly.	Home → search → character profile.
Series-focused fan	Browse a specific series.	Series page → filters → character list.
Lore explorer	Understand powers and connections.	Profile → abilities/forms → related characters.
Comparison user	Compare documented attributes.	Compare page → select characters → view differences.
Administrator/editor	Review facts and images.	Admin queue → inspect evidence/rights → approve or reject.

4. Site Map and Features

Page / feature	Planned functionality
Home / Universe Hub	Series shortcuts, featured characters, recently added profiles, popular categories and random character.
Characters Explorer	Search by name/alias; filter by series, role, species and abilities; sort and paginate.
Series pages	Overview and character list for each of the three included series.
Character profile	Summary, aliases, species, homeworld, powers, weaknesses, appearances, forms, relationships, sources and gallery.
Alien database	Browse alien characters/species and relevant transformation records.
Villains archive	Antagonist profiles, motivations, affiliations, abilities and appearances.
Gallery / forms	Rights-reviewed images, alternate forms and series-specific designs.
Character comparison	Side-by-side comparison of sourced attributes; avoid declaring an unsupported universal winner.
Admin dashboard	Research queue, image review, publishing, revisions, error logs and takedown handling.
Legal / contact	Unofficial fan-project notice, contact route, corrections and image-removal requests.

5. Visual and UX Direction

Design direction: a dark, futuristic interface inspired by the general Omnitrix sci-fi aesthetic, without copying official logos or an existing site.

Token	Value / usage
Primary accent	#A3FF12 — key actions, active states and selected highlights.
Page background	#0B0F0C — main dark canvas.
Surface	#101512 — cards, panels and navigation.
Elevated surface	#1A211C — hover states and nested panels.
Primary text	#E9EFE9 — headings and main copy.
Secondary text	#AAB5AD — metadata and descriptions.

- Mobile-first responsive layout with accessible contrast and readable typography.
- Character artwork should be prominent, but card layouts must remain fast and consistent.
- Use restrained green glow and subtle transitions; avoid visual clutter.

- Provide loading, empty, broken-image and error states.
- Use descriptive alt text and keyboard-accessible controls.

6. Recommended Technology Stack

Layer	Recommendation	Purpose
Frontend and routing	Next.js App Router + TypeScript	Public pages, routing, metadata and server-rendered content.
Styling	Tailwind CSS	Reusable responsive design system.
Database	Supabase PostgreSQL	Relational content, relationships, source records and job metadata.
Authentication	Supabase Auth	Admin sign-in and role-controlled editorial access.
File storage	Supabase Storage or approved external image hosting	Store only assets that can lawfully be hosted; retain attribution and license evidence.
AI worker	Separate TypeScript background worker	Research, structured extraction, validation, deduplication and ingestion.
Search	PostgreSQL indexes and full-text search initially	Fast name, alias and description search; external search engine only if needed later.
Testing	TypeScript checks + unit/integration/e2e tests	Catch data, permission, rendering and workflow errors.
Deployment	Next.js-compatible host + managed Supabase	Public website, database and operational management.

The AI provider and search/image discovery providers should be selected after checking current API access, pricing, rate limits, commercial terms and image-source availability. Keep these providers behind small adapters so one can be replaced without changing the database or UI.

7. High-Level Architecture

Browser → Next.js pages and API route handlers → validated service layer → Supabase PostgreSQL. A separate background worker processes research jobs and writes draft data through a restricted service interface. Admin review gates publication. Public pages read only published, approved records.

- Frontend: homepage, series pages, search, profiles, galleries and comparison UI.
- Backend: input validation, server-side access checks, rate limiting, data reads/writes and moderation actions.
- Database: canonical character records, series appearances, forms, abilities, image metadata, source evidence and audit history.
- AI worker: queue consumer, search adapters, source extraction, structured-output validation, duplicate detection and retry handling.
- Admin tools: review facts, inspect image provenance, approve/reject assets, publish records and revert edits.

8. Suggested Repository Structure

```
ben10-universe/
■■■ public/
■   ■■■ icons/
■   ■■■ placeholders/
■■■ src/
■   ■■■ app/
■   ■   ■■■ page.tsx
■   ■   ■■■ characters/page.tsx
■   ■   ■■■ characters/[slug]/page.tsx
■   ■   ■■■ series/[slug]/page.tsx
■   ■   ■■■ aliens/page.tsx
■   ■   ■■■ villains/page.tsx
■   ■   ■■■ compare/page.tsx
■   ■   ■■■ admin/
■   ■   ■■■ api/
■   ■■■ components/
■   ■■■ lib/
■   ■   ■■■ supabase/
■   ■   ■■■ validation/
■   ■   ■■■ search/
■   ■■■ types/
■■■ worker/
■   ■■■ discovery/
■   ■■■ research/
■   ■■■ image-review/
■   ■■■ deduplication/
■   ■■■ publishing/
■■■ supabase/
■   ■■■ migrations/
■   ■■■ seed/
■■■ tests/
■■■ .env.example
■■■ package.json
```

9. Database Design

Use normalized relational tables for entities and many-to-many relationships. Keep continuity-specific descriptions in the series-appearance table so that a returning character's portrayal in different included series can be documented without merging distinct versions incorrectly.

Table	Key fields and purpose
series	id, name, slug, continuity, release_year; canonical list of the three included series.
characters	id, name, slug, species_id, homeworld, summary, status, created_at, updated_at; core identity.
character_aliases	id, character_id, alias, normalized_alias; alternate names and search matching.
character_series	character_id, series_id, version_name, description, first_appearance; series-specific data.
species	id, name, homeworld, description; species records.
abilities	id, character_id, type, name, description, limitations; powers, weaknesses and limitations.
character_forms	id, base_character_id, form_character_id, form_type, notes; transformations and Ultimate forms.
character_relationships	source_character_id, target_character_id, relationship_type, evidence_source_id; documented relationships.
character_images	id, character_id, image_url, source_url, creator_or_rights_holder, license, license_evidence_url, attribution_text, review_status, alt_text, position.
sources	id, url, title, publisher, source_type, accessed_at, reliability_notes; research sources.
fact_sources	id, entity_type, entity_id, fact_key, source_id, supporting_excerpt, confidence; evidence for individual facts.
research_jobs	id, entity_name, series_id, status, attempts, error_message, started_at, finished_at; queue and monitoring.

Table	Key fields and purpose
content_revisions	id, entity_type, entity_id, old_data, new_data, reviewed_by, created_at; audit and rollback support.
admin_profiles	user_id, role, created_at; admin/editor roles mapped to authenticated accounts.

Data modelling rules

- Use UUID primary keys, foreign keys, unique slugs and timestamps.
- Use a join table for characters appearing in multiple series.
- Use explicit relationships for variants and forms; avoid duplicate records for simple aliases.
- Keep canon facts separate from speculation and mark disputed/uncertain information.
- Do not use an empty weakness list to imply that a character has no weaknesses; store unknown status separately where useful.
- Require evidence links for publishable factual claims and store image rights evidence separately from ordinary fact sources.
- Index slugs, normalized names, aliases, series relationships and common filters.

10. AI Research and Image Workflow

Stage 1 — Discovery

1. Start from a curated seed roster for each included series.
2. Search for candidate characters, aliases, species, forms and appearances.
3. Normalize names and compare against existing records to prevent duplicates.
4. Create a queued draft job; do not publish newly discovered entities automatically.

Stage 2 — Fact research

1. Collect sources for identity, species, homeworld, powers, weaknesses and appearances.
2. Prefer authoritative sources or reputable, well-documented reference sources.
3. Attach evidence to each important fact instead of assigning one citation to an entire profile.
4. Resolve conflicts where possible; otherwise flag the field for editorial review.
5. Use strict structured output and validate against a schema before saving.
6. Leave unknown details empty/unknown instead of guessing.

Stage 3 — Image discovery and rights checks

1. Search for relevant image candidates using the character name, series and desired view.
2. Capture the original source URL, creator/rights holder if known, license, license evidence, attribution requirements and review date.
3. Check that the image depicts the correct character/version and is not a duplicate or unrelated result.
4. Verify permission and license conditions before publishing or copying the asset to storage.
5. Keep unclear or incompatible rights in pending/rejected states; do not treat public availability as permission.
6. Aim for 3–4 gallery images per character, but publish fewer if there are not enough suitable, rights-cleared assets.
7. Provide a takedown/replacement workflow for complaints and broken source links.

Stage 4 — Deduplication and validation

1. Compare normalized names, aliases, series and known forms.

2. Validate URLs and required fields; detect empty or malformed content.
3. Check image duplicates using source URL and, where permitted, perceptual hashes.
4. Require a source and confidence threshold for important factual fields.
5. Route low-confidence facts, conflicting sources and unclear rights to human review.

Stage 5 — Human approval and publishing

1. Admin reviews the draft facts and source evidence.
2. Admin reviews each image's provenance and rights status.
3. Approved content changes to published status and becomes publicly visible.
4. Store revision history and allow rollback/unpublishing.
5. Schedule periodic checks for broken links, stale information and rights changes.

11. AI Job Statuses and Failure Handling

Status	Meaning / next action
queued	Waiting for a worker.
researching	Collecting facts and source evidence.
image_review	Finding and assessing image candidates.
needs_review	Human review required.
approved	Passed editorial and rights checks; ready for publication.
published	Approved and visible publicly.
failed	Worker error; inspect log and retry safely.
rejected	Rejected by editor or failed a required gate.

- Use bounded retries with backoff for temporary provider errors.
- Use idempotent job IDs so retrying a task does not create duplicate records.
- Record error codes and safe diagnostics; never log secrets.
- Pause or throttle workers when search APIs enforce rate limits.
- Provide a manual requeue option for failed jobs.

12. Security, Privacy and Content Governance

- Public users can read published content only; draft records and moderation notes must remain private.
- Use PostgreSQL Row Level Security and explicit grants. Test policies with anonymous, authenticated and admin roles.
- Keep service-role/database secret keys and AI provider keys server-side; never expose them in browser code.
- Protect admin endpoints with authentication, role checks, input validation and rate limits.
- Validate uploaded files and external URLs; avoid server-side request forgery risks when fetching external content.
- Keep an audit trail of approvals, edits, unpublishing and takedown requests.
- Publish an unofficial fan-project disclaimer, contact route, privacy notice appropriate to actual data collection, and content-removal process.
- A disclaimer does not grant rights to copyrighted characters or artwork. Obtain permission or use assets with documented reuse rights; seek legal advice for uncertain cases.

13. Search, SEO and Performance

- Create stable routes such as /characters/heatblast and /series/ben-10-2005.
- Add unique metadata, canonical URLs, Open Graph previews and sitemap entries for published pages.
- Use descriptive alt text; do not keyword-stuff names or claims.
- Implement PostgreSQL full-text search and indexes initially; evaluate a dedicated search service only if scale requires it.
- Use responsive images, lazy loading, pagination and caching for public pages.
- Add monitoring for API failures, slow queries, broken images and worker errors.

14. Testing Plan

Test area	Acceptance checks
Unit tests	Schema validation, slug normalization, duplicate matching and rights-state transitions.
Integration tests	Database relationships, API validation, publishing flow and worker retries.
Access control	Anonymous users cannot read drafts or execute admin actions; non-admins cannot approve.
UI / end-to-end	Search, filters, profile navigation, gallery fallbacks and mobile layouts.
AI quality	Unknown facts remain unknown; sources are attached to claims; fabricated URLs are rejected.
Image rights	Assets without acceptable evidence cannot be published.
Reliability	Duplicate jobs are safe to retry; failed jobs are visible and recoverable.
SEO	Published pages have metadata and sitemap entries; drafts are not indexed.

15. Implementation Roadmap

Phase	Work	Deliverable / exit condition
1. Foundation	Create app, design tokens, routing, Supabase project, environment setup.	Responsive site shell and secure database connection.
2. Data model	Migrations, seed series, character records and source tables.	One fully working profile backed by real data.
3. Core UX	Home, series lists, search, filters, profile, gallery and villain archive.	Users can browse the initial verified roster.
4. AI research	Discovery, research adapters, structured outputs, citations and deduplication.	AI creates reviewable drafts with evidence.
5. Image review	Candidate gallery, license evidence, moderation and rejection flow.	No image is published without passing rights checks.
6. Admin/publishing	Auth roles, review queue, revisions, logs and publish/unpublish.	End-to-end controlled content pipeline.
7. Launch	SEO, accessibility, performance, backup, security and regression tests.	Public beta ready for ongoing catalog growth.

Do not estimate a firm launch date until the selected AI/search providers, hosting and initial content workload are known. Complete the architecture and a small vertical slice first, then estimate based on measured progress.

16. Initial Release Acceptance Checklist

- ■ Only Ben 10 (2005), Alien Force and Ultimate Alien are in the planned series scope.
- ■ Omniverse and the 2016 reboot are excluded from navigation and content seeding.
- ■ Search and filters work across names, aliases, series and roles.
- ■ Profiles show source-backed facts and appropriate unknown states.
- ■ Gallery images include source and rights metadata; unapproved images are not public.
- ■ Admin access and row-level permissions are tested.
- ■ Character forms and series-specific versions are modelled correctly.
- ■ Broken-image, empty-gallery and failed-worker states are handled.
- ■ Published pages have SEO metadata and sitemap entries.
- ■ A content-removal process and unofficial fan-project notice are visible.

17. Risks and Mitigations

Risk	Mitigation
Incorrect AI-generated lore	Fact-level citations, confidence checks, unknown values and editor approval.
Copyright or licensing problems	Documented rights review, conservative publication gates and takedown process.
Duplicate/incorrect variants	Continuity-aware schema, alias normalization and human review of uncertain matches.
API cost or rate limits	Provider adapters, quotas, caching, queue throttling and batch jobs.
Broken external images	Periodic link checks, fallback images and preference for rights-cleared stable hosting.
Unauthorized content changes	Server-side authorization, RLS, audit history and tested permissions.

18. Recommended Build Order

1. Create the repository, Next.js app and reusable design system.
2. Set up Supabase and implement migrations for series, characters, sources and images.
3. Build one complete character profile using verified starter data.
4. Build series browsing, search, filters and villain archive.
5. Add secure admin authentication and manual content review first.
6. Implement AI discovery and fact research in draft-only mode.
7. Add image discovery and rights evidence checks.
8. Test the full pipeline on a small set of characters before scaling up.
9. Deploy the public beta and expand the catalog gradually.

19. Final Project Definition

The first release is a polished, public, unofficial Ben 10 character encyclopedia covering exactly three series: Ben 10 (2005), Ben 10: Alien Force and Ben 10: Ultimate Alien. It includes character and villain profiles, alien forms, powers, weaknesses, sourced lore, galleries, search, filtering and an admin dashboard. A separate AI worker prepares content and image candidates, while source evidence, usage-rights checks and approval gates protect the quality of published material.

The first coding milestone is a working website shell, the initial database migrations and one complete source-backed character profile. The AI agent should be connected only after this foundation works reliably.